

RESEARCH ARTICLE

How Do Root Morphological Traits Regulate Solute Transport? Evidences From Forest Ecosystems, Eastern China

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ABSTRACT

The heterogeneity of solute transport is potentially the result of spatial variations in root morphological traits across different diameter classes. However, the mechanism of root systems on solute transport remains unclear. In this study, soil columns collected from *Quercus acutissima*-, *Pinus taeda*- and *Phyllostachys edulis* stands were used to investigate the relationship between solute transport and root systems. The results showed that solute transport parameters like the average pore water velocity (V), dispersion coefficient (D) and distribution coefficient (β) decreased with increasing soil depth, while the dimensionless mass transfer coefficient (ω) showed fluctuating changes. The V , D and β in topsoil (0–20 cm) were 1.39–57.41 times larger than those in subsoil (30–50 cm), while the ω in subsoil was 2.30–2.36 times larger than that in topsoil. The RLD of fine roots (diameter: 0–1 mm) and the total RSAD could significantly promote the V . There was a positive relationship between the D and RSAD of medium roots (diameter: 1–3 mm) and RLD of total roots. The RLD of medium roots and total roots could both have positive effects on β . Yet, the negative effects of the RSAD and RVD of medium roots on ω were observed. *Quercus acutissima* stands had strong soil and water conservation ability but increased the risk of retaining pollution. *Pinus taeda* stands could promote rapid infiltration and resist pollution retention. Although *Phyllostachys edulis* stands had weak soil and water conservation ability, they reduced the risk of pollution downward transport. This study advanced the understanding of the potential regulation on solute transport and provided suggestions for forest management.

1 | Introduction

Forest ecosystems play an important role in the hydrological cycle (Kuehler et al. 2017; Xie et al. 2020). They facilitate efficient soil infiltration, regulate groundwater recharge and enhance soil and water conservation (Yang and Zhang 2011; Kuehler et al. 2017). However, unsustainable forest management practices, like the conversion of forest stands to agricultural lands, would lead to the transport of chemical fertilisers and pesticides from agricultural production into the groundwater through the

vadose zone (Wang et al. 2013), which results in groundwater pollution and further impairs forest health and hydrological functions (Tang et al. 2024a, 2024b). Therefore, it is necessary to adequately understand solute transport in forest ecosystems and accordingly develop sustainable forest management practices.

Solute transport is an important process in hydrological cycles (Vogel et al. 2006; Zhang et al. 2022b, 2023). For example, the extensive use of chemical fertilisers and pesticides to meet the grain demands would cause severe soil and groundwater

pollution by solute transport in the soil (Zhang et al. 2015). Thus, the mechanisms of the solute transport require more attention to reduce the possible negative effects of this behaviour in hydrological cycles (Cremer et al. 2016; Zhang et al. 2023; Ju et al. 2025). The control of pollution needs accurate prediction of pollution transport. Previous studies have tried several practices to detect the solute transport characteristics. A well-recognised method used to reflect the solute transport characteristics in soil column was the breakthrough curves (BTCs) of conservative tracers (Yang et al. 2013). This is because the related parameters inversely estimated from BTC could well reveal the solute transport process (Zhang et al. 2015). In addition, the study of solute transport behaviour at the soil column scale plays an important role in the prediction of eco-hydrological responses at larger scales (Koestel et al. 2013). Yet, many models have been developed to fit BTC. Compared with the traditional convection dispersion model (CDE), the two-region transport model (TRM) additionally considered the mobile-immobile regions in the soil column, which was more suitable to fit the BTC in the well-developed soil structure (Zhang et al. 2020a, 2020b; Zaheer et al. 2024). This is because solutes preferentially pass in the 'mobile region' while passing with the negligible flow velocity in the 'immobile region', which was more in line with the actual solute transport in forest soils (Rezanezhad et al. 2016; Zhang et al. 2020a, 2020b). For example, Zhang et al. (2015) found that the TRM improved the simulation performance of a specific soil column in the Sunjia agricultural catchment compared to CDE. Guleria et al. (2019) showed that the TRM exhibited high precision for the simulation of BTC in a layered soil porous media. Thus, in this study, the TRM was used to fit the BTC.

Various factors, such as soil properties, plant root systems and soil fauna, would affect the solute transport (Tang et al. 2024b; Ju et al. 2025). Root systems have significant effects on solute transport in the soil by increasing soil porosity and thereby reducing the soil bulk density (Leung et al. 2018). This is because, on the one hand, the activities of roots can loosen the soil and facilitate the hydrologically active (Xie et al. 2020). On the other hand, biological gaps and macropores between root systems and the surrounding soil formed by root decay would create preferential flow paths (Tang et al. 2024b). However, the impact of different diameter classes of root systems and their morphological traits on solute transport might vary (Chen et al. 2025) and the opposite results might be obtained. For example, Zhang et al. (2022b) found that coarse root systems showed negative effects on the rapid solute transport due to the formation of 'water accumulation' around the root surface, while Guo et al. (2020) concluded that coarse roots would facilitate rapid solute transport by promoting preferential flow. In addition, different root morphological traits show various contributions to the solute transport. This is because root length is related to the formation of root channels, root surface area could affect the interaction along the root-soil interface and root volume changes the soil structure by squeezing soil particles (Tang et al. 2024b; Cai et al. 2024). Therefore, the effects of root morphological traits on specific solute transport parameters might be different. Yet, studies on the influence of different root diameter classes and their morphological traits on solute transport were extremely rare. Especially for forest ecosystems, it is characterised by highly developed root systems (Xie et al. 2020). It is recognised that different forest types showed high heterogeneity of root

morphology traits and spatial distribution. Investigating the properties of root systems in different forest stands and detecting their influences on solute transport are conducive to enhancing the mechanistic understanding of plant-soil interactions (Xie et al. 2020). Nonetheless, to the best of our knowledge, the mechanism by which diameter classes of the root morphological traits at different forest stands influence solute transport was rarely studied. Could each root morphological trait affect solute transport? And in further, which one of the forest stands is suitable for soil and water conservation or reducing pollution downward transport? It remains unclear.

In summary, previous studies have indicated that soil structure spatial variation-induced solute transport differs (Zhang et al. 2015). However, the relationship between solute transport parameters and root systems has rarely been concluded. We hypothesised that various diameter classes of root systems and their morphological traits contribute differently to the solute transport parameters. The specific objectives of this study were to (1) characterise solute transport and root morphological traits in different soil depths in forest stands, (2) find the interaction between the characteristics of root morphological traits with different diameter classes and solute transport and (3) propose some implications for forest management according to the results. Assessing the effects of root systems on solute transport in the soil could better our understanding of the effects of soil biological factors in solute transport, which can offer scientific help for sustainable forestry management.

2 | Materials and Methods

2.1 | Study Areas Determination

The study area (Yangtze River Delta Forest Ecological Station) is in Xiashu Town, Jurong City, Jiangsu Province, eastern China (32°07' N, 119°13' E) (Figure 1a). This region has a typical northern subtropical monsoon climate with a mean annual rainfall of 1055.6 mm and an annual mean temperature of 15.2°C. The study area has an area of about 314.4 ha, with the coverage of the forest stand > 90% (Chen et al. 2025). The soils in the forest stands are characterised by silty loam texture (Tang et al. 2024b). The thickness of the soil is about 50 cm, and below the soil depth of 50 cm is the parent material layer (Chen et al. 2025).

The main forest stands in the study area include oak, pine and bamboo stands (Tang et al. 2024a, 2024b). The main tree species in oak stands are *Quercus acutissima* and *Quercus variabilis*, in pine stands are *Pinus taeda*, *Pinus massoniana* and *Pinus elliottii* and in bamboo stands are *Phyllostachys edulis* and *Phyllostachys viridis*. In this study (Figure 1b), *Q. acutissima* stands, *P. taeda* stands and *P. edulis* stands were selected because they have wide distribution areas and similar growth history (from the 1970s to 2025) and play important roles in soil and water conservation in the local area (Tang et al. 2024a). The soils basic physical and chemical properties in these stands were provided in Table 1. In addition, farmlands are around the selected stands. Farmers apply pesticides and fertilisers to combat pests and increase the yield. Those foreign additives could affect natural stands through solute transport in the soil (Tang et al. 2024b). Therefore, the study of solute transport can provide assistance in

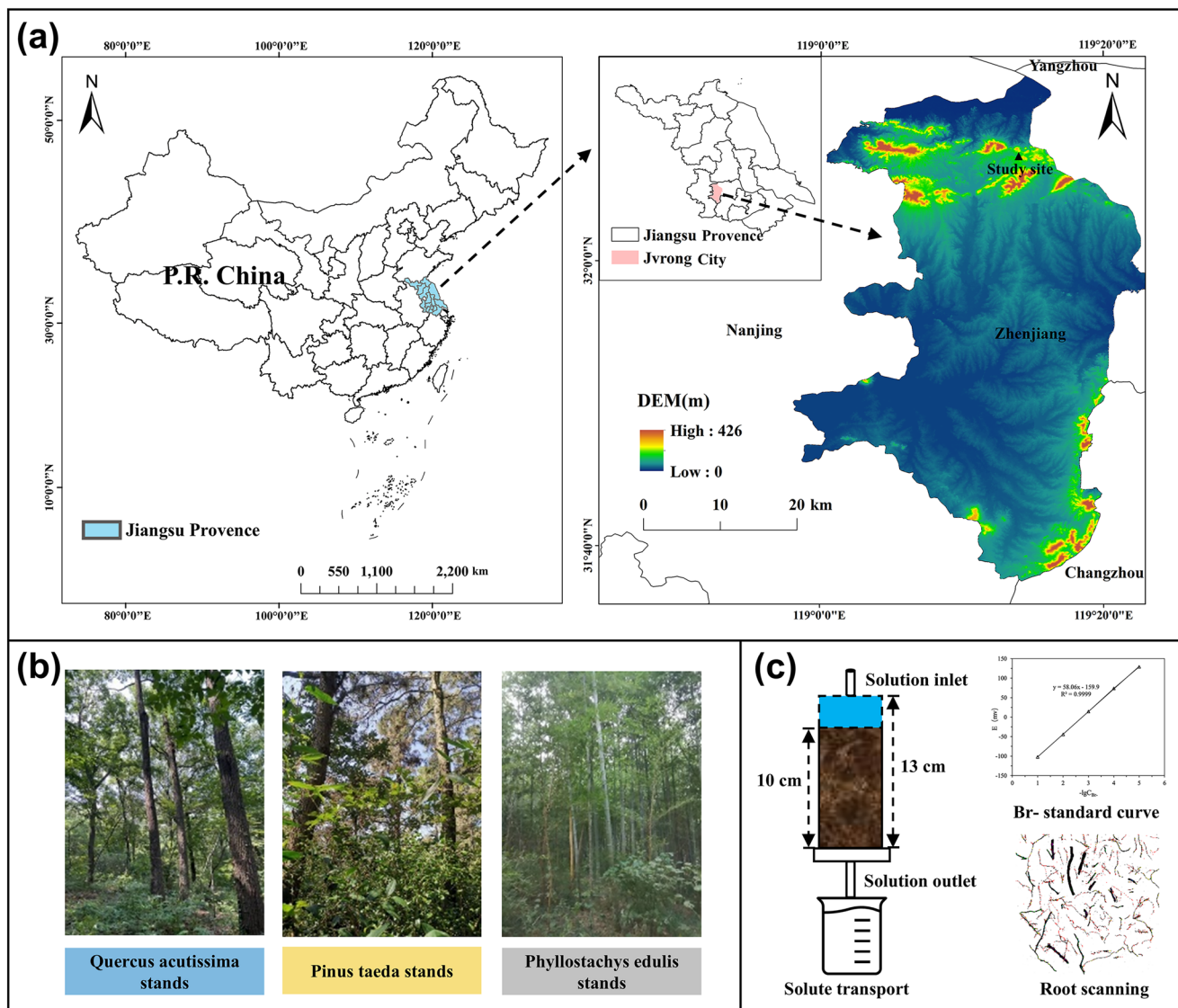


FIGURE 1 | (a) Location of the study site in eastern China. (b) Samples of the three forest stands, including *Q. acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands. (c) Left panel: 10 cm soil cores solution transport experimental set up. Water head was 3 cm. At the outlet, the columns were equipped with a sample collector and an ion meter (PXSJ-270F, Shanghai Yidian Scientific Instrument Co. China) to measure the concentrations of all the leachates. Right panel: Br⁻ standard curve ($R^2 > 0.99$) (above). The root morphological traits were analysed by WinRHIZO 2016a (Rengent Instruments Inc., Canada) (below).

assessing the degree of the foreign threat to forest stands (Zhang et al. 2022a).

2.2 | Soil Sampling Methods

In October 2021, three experimental sites (10 m × 10 m) were established under similar site conditions at each forest stand. The experimental sites were independent of each other. Three replicate plots were created at each site. Before sampling, the litter layer and gravel on the soil surface should be carefully removed. Then, soil profiles (50 cm in depth and 100 cm in width) were excavated to collect soil cores. A cutting cylinder (10 cm in height and 7.2 cm in diameter, 400 cm³) was used to take soil columns. Each cutting cylinder was sealed with plastic film to reduce the evaporation of soil moisture. The same method was used to take soil columns at the soil depth of each 10 cm. A total of

soil columns at five soil depths (0–10, 10–20, 20–30, 30–40 and 40–50 cm) were collected. Thus, a total of 135 undisturbed soil columns (3 forest stands × 3 sites × 3 reduplicates × 5 soil depths) were collected in this study.

2.3 | Solute Transport Experiments

The bromide was chosen as the tracer of the tracer BTC experiments because it was a relatively conservative tracer with low adsorption ability (Zhang et al. 2015; Alessandrino et al. 2022; Zhou et al. 2023). Therefore, the 0.05 mol/L NaBr solution was prepared. The closed pores on the surface of the soil column should first be exposed to restore its natural structure. Then, the soil column was placed in deionised water for 24 h to make it fully saturated (Zhang et al. 2015; Alessandrino et al. 2022). After the soil column was fully saturated, it was placed in dry

TABLE 1 | Changes of soil chemical and physical properties across the entire soil profile (0–50 cm) in the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands). Soil chemical properties included soil organic carbon (SOC), soil total nitrogen (STN), soil total phosphorus (STP), soil total potassium (STK) and pH. Soil physical properties included soil texture (clay, silt and sand content), soil bulk density (SBD), total soil porosity (TSP) and initial soil water content (ISWC).

Indicators		Unit	Forest stands		
			<i>Quercus acutissima</i> stands	<i>Pinus taeda</i> stands	<i>Phyllostachys edulis</i> stands
Soil chemical properties	SOC	g/kg	6.50 ± 5.40	7.22 ± 5.63	5.46 ± 3.30
	STN	g/kg	0.74 ± 0.44	0.85 ± 0.41	0.68 ± 0.29
	STP	g/kg	172.73 ± 28.8	197.69 ± 47.70	153.92 ± 32.97
	STK	g/kg	13.00 ± 0.42	15.56 ± 1.36	14.82 ± 1.19
	pH		3.66 ± 0.05	4.24 ± 0.17	4.33 ± 0.16
Soil physical properties	Clay content	%	1.16 ± 0.56	1.09 ± 0.98	1.76 ± 1.11
	Silt content	%	84.73 ± 7.52	86.85 ± 5.24	79.54 ± 7.08
	Sand content	%	14.11 ± 7.80	12.05 ± 5.63	18.70 ± 7.01
	SBD	g/cm ³	1.33 ± 0.18	1.29 ± 0.19	1.26 ± 0.15
	TSP	%	41.09 ± 4.18	47.21 ± 5.68	48.90 ± 5.93
	ISWC	%	10.34 ± 1.03	25.08 ± 2.85	25.34 ± 1.99

sands for 12 h to exclude the gravity of water (Zhang et al. 2015). Filter papers were placed above and below the soil column, respectively, to prevent the direct erosion of the soil surface when adding water and reduce the loss of soil particles in the soil column (Ju et al. 2025). In addition, the Vaseline was applied at the interaction between the soil column and the wall of the cutting cylinder to prevent the water flow on the wall edge. The Mariotte bottle was used to supply the deionised water above the soil column at a constant water head of 3 cm (Kang et al. 2015; Ju et al. 2025). Then, the saturated hydraulic conductivity (K_s) was measured (Ju et al. 2025). After forming a stable water flow condition inside the soil column, the deionised water at the top of the soil column was excluded (Zaheer et al. 2024). The 0.05 mol/L NaBr solution was then put into the top of the soil column with a constant water head of 3 cm (Kang et al. 2015; Ju et al. 2025). The leachate was collected, and the concentrations of all the leachates were measured using an ion meter (PXSJ-270F, Shanghai Yidian Scientific Instrument Co. China) (Zhang et al. 2015) (Figure 1c). The collection frequency of the leachate should be fully considered (Zhang et al. 2020b; Alessandrino et al. 2022). In the initial stage and rapid replacement stage of the solute transport process, the concentration of the leachates increased rapidly. Thus, it was needed to increase the collection frequency of the leachate (Ju et al. 2025). When the solute transport process was in the complete stage, the concentration of the leachate tended to stabilise, and the frequency of leachate collection could be reduced (Ju et al. 2025). Therefore, the application of solute volume, duration of NaBr solution and collection frequency of the leachate were ultimately determined based on the preliminary solute transport experiments, which were specified in Table 2. The concentrations of all the leachates are expressed as relative concentration (C/C_0), which is achieved by dividing the experimental concentration (C) by the initial NaBr solution concentration (C_0) (Zaheer et al. 2024). Finally, the BTC of the solute transport in the soil was obtained. The

relative concentration approaching 1.0 could indicate that the concentration of the leachates has reached or is close to the initial NaBr solution concentration, and the experiments should stop (Zaheer et al. 2024; Ju et al. 2025) (Figure 1c).

2.4 | Transport Models Setting

The macropore of forest soils is abundant and well-structured because of the growth of the root systems and the activity of soil animals (Zhang et al. 2015). Therefore, a two-region model (TRM) was introduced to better characterise the solute transported in heterogeneous forest soils compared with the traditional CDE (Van Genuchten and Wierenga 1976).

$$R \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial z^2} - v \frac{\partial C}{\partial z} \quad (1)$$

where C is the solute concentration (mol/L), t is the time (min), z is the distance in the vertical direction (cm), D is the solute dispersion coefficient (cm²/min) and V is the average pore water velocity (cm/min).

In the TRM, water-filled pore space could be divided into mobile and immobile regions. The mobile region dominates the convective–dispersive transport, while the presence of the solute in the immobile region is dependent on diffusion from the mobile to the immobile region (Van Genuchten and Wierenga 1976; Kang et al. 2015; Zaheer et al. 2024):

$$\theta_m \frac{\partial C_m}{\partial t} + \theta_{im} \frac{\partial C_{im}}{\partial t} = \theta_m D \frac{\partial^2 C_m}{\partial x^2} - v_m \theta_m \frac{\partial C_m}{\partial x} \quad (2)$$

$$\theta_{im} \frac{\partial C_{im}}{\partial t} = \omega (C_m - C_{im}) \quad (3)$$

TABLE 2 | The application volume, duration of NaBr solution, and collection frequency of the leachate in solute transport with soil depth increasing in the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands).

Forest stands	Soil depths (cm)	Application of solute volume (ml)	Duration of NaBr solution (min)	Collection frequency of the leachate		
				Initial stage (min)	Rapid replacement stage (min)	Complete stage (min)
<i>Quercus acutissima</i> stands	0–10	1500	300	2	5	10
	10–20	1000	500	3	10	15
	20–30	800	1500	10	15	60
	30–40	600	2000	10	30	60
	40–50	400	2500	20	60	100
<i>Pinus taeda</i> stands	0–10	1500	100	1	5	10
	10–20	1200	500	5	10	30
	20–30	1000	1000	5	15	30
	30–40	800	1200	5	30	60
	40–50	600	1500	5	30	120
<i>Phyllostachys edulis</i> stands	0–10	1000	500	5	10	20
	10–20	900	800	5	10	30
	20–30	800	1200	10	20	90
	30–40	700	2000	10	30	90
	40–50	600	2200	10	60	100

$$\theta = \theta_m + \theta_{im} \quad (4)$$

$$\beta = \frac{\theta_m}{\theta} \quad (5)$$

where θ is the soil volume water content (cm^3/cm^3), and θ_m and θ_{im} are the soil water content in mobile and immobile regions (cm^3/cm^3), respectively. Similarly, C_m and C_{im} are the solute concentrations in mobile and immobile regions (mol/L). v_m is average mobile pore water velocity (cm/min). β is the distribution coefficient between the mobile and the immobile regions. ω is the dimensionless mass transfer coefficient, which could control the rate of the particle exchange between the mobile and immobile regions (T^{-1}) (Toride et al. 1999).

TRM is available in CXTFIT 2.1, which is a one-dimensional solute transport model included in the STANMOD (Toride et al. 1999; Liu et al. 2023). This model estimates the solute transport parameters from the measured BTC. The least squares method was used to minimise the difference between the calculated and measured BTC (Toride et al. 1999; Ju et al. 2025). The square of the correlation coefficient (R^2), mean-square error (MSE) and squared weighted deviations (SSQ) were calculated to evaluate the model performance. The lower the SSQ and MSE, and the higher the R^2 , the better the fit between the simulated and measured values (Zhang et al. 2023; Zhou et al. 2023; Zaheer et al. 2024).

2.5 | Root Morphological Traits Acquisition

After the solute transport experiment, the intact soil column within the cutting cylinder was carefully removed. Then, the soil column was placed on a 1 mm soil sieve and washed under the running water until the soil remaining on the root surfaces was removed. The cleaned root systems were dried in the 80°C oven until surface moisture was eliminated (Zhang et al. 2022a). Next, the dried root systems were arranged against overlapping on a root dish for scanning. An EPSON 12000XL flatbed scanner at a resolution of 400 dpi was used to obtain two-dimensional grayscale images of the root systems (Figure 1c). The obtained images were then extracted using the WinRHIZO Pro root analysis system to analyse root morphological traits. In this study, root systems could be further classified into four categories, that is, fine roots ($0 < d \leq 1 \text{ mm}$), medium roots ($1 < d \leq 3 \text{ mm}$), coarse roots ($3 < d \leq 4 \text{ mm}$) and extremely coarse roots ($d > 4 \text{ mm}$) according to their different roles in ecological processes (Ye et al. 2017; Zhang et al. 2022a; Wang et al. 2024; Chen et al. 2025). This is because fine roots exhibit strong retention capacity, medium roots act in the water transport within plants, coarse roots show positive effects on water retention and very coarse roots primarily promote soil structural stability (Chen et al. 2025). The root length density (RLD), root surface area density (RSA) and root volume density (RVD) of different root categories could be calculated by dividing RL, RSA, and RV obtained from the WinRHIZO Pro

root analysis system by soil column volume (400 cm³) (Zhang et al. 2022a; Wang et al. 2024).

2.6 | Data Analysis

The normality of the data was tested for normal distribution using the Shapiro-Wilk test, and the non-normally distributed variables were log transformed. One-way analysis of variance (ANOVA) was used to compare the differences in the solute transport parameters between the three forest stands and different soil depths using the SPSS 22.0 software (Statistical Graphics Crop, Princeton, USA). The results were statistically significant at the 5% significance level. In this study, the random forest model (RFM) was constructed to detect the contribution of root morphological traits to solute transport. The RFM was chosen for its ability to efficiently capture complex interactions between predictor variables and response variables, and it requires minimal parameter tuning compared to other machine learning algorithms (Ju et al. 2024; Zhou et al. 2025). We began by optimising the hyperparameters of the RFM using Grid Search. Then, 10-fold cross-validation was conducted to further validate the performance of the model and safeguard against overfitting (Zhou et al. 2025). The significance of the RFM and the variables was analysed using the 'randomForest' and 'caret' packages. Although RFM could identify the root morphological traits that have relatively larger contribution to solute transport parameters, the specific relationships between these key traits and solute transport parameters require further investigation. The generalized additive mixed effect model (GAMM) was employed to explore the effects of the selected root morphological traits on solute transport parameters (Ziccardi et al. 2025). The selected root morphological traits were entered via linear or smoothing functions against the solute transport parameters (as dependent variable) (Yun et al. 2024). The GAMM function in the 'mgcv' package was employed to fit the smoothing terms with the default thin plate spline technique (Pedersen et al. 2019; Yun et al. 2024). What's more, the gam function used penalised regression splines of moderate rank for the smoothing function (Pedersen et al. 2019; Yun et al. 2024). To further evaluate the relationship between the selected root morphological traits and solute transport parameters in each forest stand, linear regression analysis was performed. The RFM, GAMM and linear regression analysis were conducted in R 4.4.1 (<https://www.r-project.org/>). Finally, the partial least squares path models (PLS-PM) were constructed using SmartPLS 3.2.9 software (<https://www.smartpls.com/>). These models illustrated the mechanisms by which root systems, total soil porosity and silt drive changes in solute transport parameters. The selected root morphological traits were obtained from the RFM.

3 | Results

3.1 | Solute Transport Characteristics

The fitted BTCs of solute transport were obtained by the TRM (Figure 2). The BTC in the soil column showed an S-shaped distribution. In the three stands, the concentration of Br⁻ in the percolates was close to 0 at the beginning. Then, the concentration began to increase sharply until relative concentration (C/C₀)

approached 1. Finally, the relative concentration remained stable at a value around 1. The breakthrough of Br⁻ occurred more quickly in the topsoil (0–20 cm) compared with that in the subsoil (30–50 cm). The *Q. acutissima* stands demonstrated the largest initial and entire transport time in the subsoil among the three forest stands, while the *P. edulis* stands showed a larger initial and entire transport time in the topsoil than the other two stands.

The TRM could fit the BTCs well with $R^2 > 0.99$, $MSE < 0.001$ and $SSQ < 0.01$ (Figure 2). There were significant differences in solute transport parameters at different soil depths in the three forest stands (Table 3). V and D showed a decreasing trend with increasing soil depth in the three stands, while the β and ω changed with fluctuations. The β in topsoil (0–20 cm) was larger than in subsoil (30–50 cm), indicating larger mobile regions in topsoil. However, the ω in topsoil was significantly smaller than in subsoil ($p < 0.05$). V, D, β , and ω across the entire soil profile (0–50 cm) exhibited stand-dependent changes. *Q. acutissima* stands showed the largest ω (0.44 ± 0.20) compared to the other stands. Meanwhile, V and D in *P. taeda* stands were the largest among the three forest stands. *Phyllostachys edulis* stands demonstrated the lowest V (0.07 ± 0.05 cm/min) and ω (0.15 ± 0.08) across the entire soil profile. There was no significant difference in the β values among the three forest stands ($p > 0.05$) (Table 3).

3.2 | Root Morphological Traits

Figure 3 indicated the changes of root morphological traits among different forest stands and soil depths. Results showed that total RLD, RSAD and RVD in the three forest stands decreased significantly with increasing soil depth ($p < 0.05$). The total RLD, RSAD and RVD in *P. edulis* stands were significantly larger than those in *Q. acutissima* and *P. taeda* stands ($p < 0.05$), while the differences of these traits in *Q. acutissima* and *P. taeda* stands were not significant ($p > 0.05$).

As shown in Figure 3, the RLD and RSAD of the fine roots ($0 < d \leq 1$ mm) constituted the main part of total RLD and RSAD, with the contribution of 87.27% of total RLD and 60.37% of total RSAD. However, the medium roots ($1 < d \leq 3$ mm) exhibited relatively larger RVD compared with the roots of other diameter classes, accounting for 48.49% of total RVD. Comparison of the root morphological traits in various forest stands revealed some differences between them. In *Q. acutissima* stands, the contribution of extremely coarse roots ($d > 4$ mm) to total root morphological traits (RLD, RSAD, and RVD) decreased with increasing soil depths, while it increased at first and then decreased in *P. taeda* stands. Yet, the proportion of total root morphological traits (RLD, RSAD and RVD) contributed by extremely coarse roots was $< 1\%$ in *P. edulis* stands.

3.3 | Interactions Between Root Morphological Traits and Solute Transport Parameters

Root morphological traits could affect the solute transport parameters (Figures 4, 5, and 6). Results from RFM showed that root morphological traits could explain 60.3% of the variation

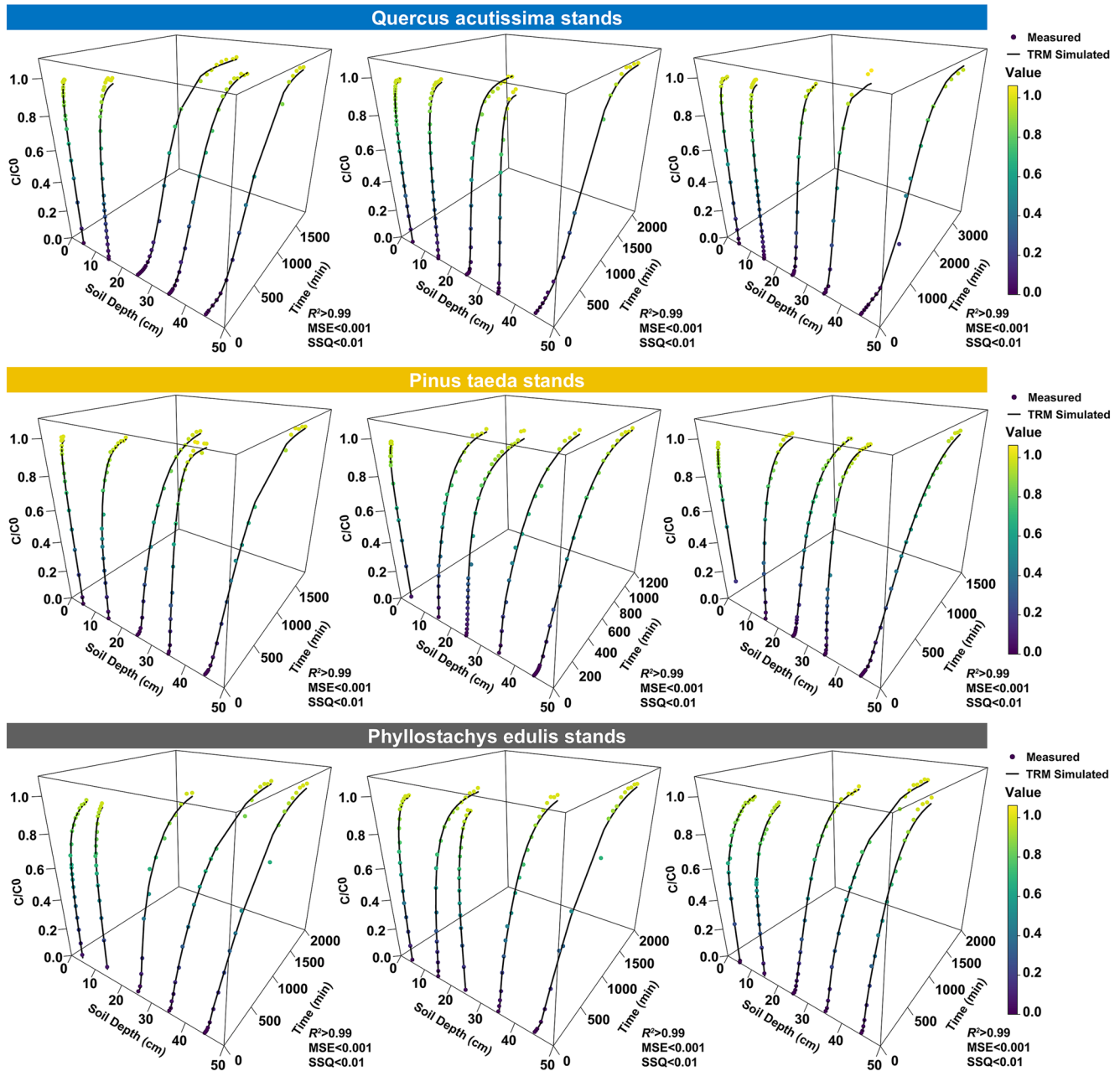


FIGURE 2 | Observed and fitted breakthrough curves (BTC) using a two-region model (TRM) for the soil columns in different soil depths (0–10, 10–20, 20–30, 30–40 and 40–50 cm) in the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands). The square of the correlation coefficient (R^2) > 0.99, the mean-square error (MSE) < 0.001 and squared weighted deviations (SSQ) < 0.01.

in the V, 70.0% in the D, 45.2% in the β , and 52.5% in the ω (Figure 4). The RLD of fine roots ($0 < d \leq 1$ mm) was an important factor influencing the V and β , and the RSAD of medium roots ($1 < d \leq 3$ mm) was an important factor influencing the D and ω . In addition, the RLD of total roots plays important roles in the D and β , and the RSAD of total roots and the RVD of medium roots ($1 < d \leq 3$ mm) have great influence on the V and ω , respectively. The GAMM revealed distinct patterns of variation in the solute transport parameters in relation to the selected root morphological traits (Figure 5). Both the RSAD of the total roots and the RLD of fine roots ($0 < d \leq 1$ cm) positively affected the V, which could be indicated in all three stands ($p < 0.05$) (Figure 6). Similarly, the larger D, the larger the RSAD of medium roots ($1 < d \leq 3$ mm) and the RLD of total roots was observed (Figure 5).

However, there was substantial variability between stands, with no significant positive relationship between root morphological traits and the D in *P. taeda* stands (Figure 6). The GAMM could not well catch the variation of the β with the deviation explanation rate of 27.5% (Figure 5). Meanwhile, there was no significant linear relationship between root morphological traits and the D in the three forest stands ($p > 0.05$) (Figure 6), demonstrating that root morphological traits were not the main effects on the β . The ω exhibited a notably negative relationship with the RVD of medium roots ($p < 0.05$) (Figures 5 and 6), while the relationship was not significant between the RSAD of medium roots and the ω in the three forest stands ($p > 0.05$) (Figure 5). These GAMMs provided an integrated analysis of the partial effects of the selected root morphological traits, confirming

TABLE 3 | Changes in solute transport parameters with soil depth in the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands).

Forest stands	Soil depths (cm)	ν (cm/min)	D (cm ² /min)	β	ω
<i>Quercus acutissima</i> stands	0–10	0.37 ± 0.17 Ba	0.59 ± 0.10 Aa	0.49 ± 0.24 Aa	0.18 ± 0.05 Ab
	10–20	0.17 ± 0.01 Ab	0.21 ± 0.09 Bb	0.35 ± 0.09 Ba	0.29 ± 0.15 Ab
	20–30	0.03 ± 0.01 Ab	0.02 ± 0.02 Ac	0.43 ± 0.27 Aa	0.49 ± 0.08 Aa
	30–40	0.05 ± 0.05 Ab	0.04 ± 0.06 Ac	0.16 ± 0.08 Ba	0.68 ± 0.13 Aa
	40–50	0.01 ± 0.01 Ab	0.01 ± 0.01 Bc	0.37 ± 0.41 Aa	0.57 ± 0.08 Aa
<i>Pinus taeda</i> stands	0–10	1.05 ± 0.45 Aa	2.43 ± 2.37 Aa	0.51 ± 0.06 Aa	0.07 ± 0.06 Bc
	10–20	0.10 ± 0.02 Bb	0.13 ± 0.06 Bb	0.33 ± 0.04 Bb	0.14 ± 0.04 Abc
	20–30	0.05 ± 0.01 Ab	0.25 ± 0.26 Ab	0.50 ± 0.11 Aa	0.24 ± 0.14 Bab
	30–40	0.07 ± 0.02 Ab	0.03 ± 0.02 Ab	0.28 ± 0.05 Ab	0.20 ± 0.05 Babc
	40–50	0.02 ± 0.01 Ab	0.01 ± 0.01 Ab	0.32 ± 0.03 Ab	0.30 ± 0.03 Ba
<i>Phyllostachys edulis</i> stands	0–10	0.15 ± 0.05 Ba	0.32 ± 0.18 Aa	0.33 ± 0.05 Ab	0.08 ± 0.03 Bc
	10–20	0.07 ± 0.03 Bab	0.46 ± 0.10 Aab	0.65 ± 0.21 Aa	0.13 ± 0.13 Abc
	20–30	0.09 ± 0.10 Aab	0.21 ± 0.32 Aab	0.40 ± 0.12 Ab	0.07 ± 0.03 Bc
	30–40	0.03 ± 0.01 Ab	0.04 ± 0.01 Ab	0.34 ± 0.06 Ab	0.21 ± 0.05 Bab
	40–50	0.02 ± 0.01 Ab	0.01 ± 0.01 ABb	0.22 ± 0.07 Ab	0.26 ± 0.03 Ba

Note: Capital letters represent significant differences in different forest stands at the same soil depth and lowercase letters represent significant differences in the same forest stand at the different soil depths ($p < 0.05$).

the patterns observed in the linear regression analysis between root morphological traits and solute transport parameters. The results of PLS-PM verified that root systems, total soil porosity and silt provided substantial positive contributions for V , D and β (total effects: 0.235, 0.284 and 0.256, respectively), while negative contributions for the ω (total effects: -0.707). Further, the root systems mainly affect the solute transport parameters by influencing total soil porosity indirectly (Figure 7).

4 | Discussion

4.1 | Effects of Root Morphological Traits on Solute Transport

In this study, the bromide BTC curves showed in the form of an S-shape pattern with no inflection points or delays (Figure 2), which was inconsistent with Luo et al. (2024). This is because the tracer solute replaced the deionised water during the initial stage of penetration (Ju et al. 2025). Then, in the rapid replacement stage, the concentration of bromide in the leachate gradually increased under the effect of convective dispersion (Zaheer et al. 2024). Finally, the concentration of bromide in the leachate approached the tracer solute in the complete stage. It was noteworthy that the initial and entire transport time of the BTC curves increased with soil depths increasing (Figure 2). In consistent with Zaheer et al. (2024), the earlier arrival of the BTC in the topsoil (0–20 cm) than that in subsoil (30–50 cm) probably reflects the stronger effects of preferential flow (Larsbo et al. 2014; Ju et al. 2025). This is because the topsoil showed a larger content of macropores and greater pore connectivity compared with the subsoil (Tang et al. 2024b), which could promote the formation

of preferential flow paths (Zhang et al. 2021b). Further, the forming specific soil water flow paths allowed the tracer to move at a rapid rate without being absorbed into the soil matrix (Zhang et al. 2021b). Therefore, the BTC for the topsoil with more soil macropore domain showed a larger concentration and faster arrival time compared to the subsoil (Köhne et al. 2009; Zhang et al. 2015). In addition, the early time tailing in subsoil was observed compared with that in topsoil, indicating that there was less tracer transport between the mobile-immobile region in topsoil than in subsoil (Xu et al. 2022). Therefore, the mobile regions where the advective transport developed in topsoil were stronger than in subsoil (Stults et al. 2022). On the contrary, the late-time flattening of the BTC curves could be attributed to the dispersion of the solute into the porous medium (Zaheer et al. 2024).

In agreement with our hypothesis, different diameter classes of root systems and their morphological traits contribute to the solute transport parameters (V , D , β and ω) differently (Figures 5 and 6). We found that the RLD of fine roots, RSAD and RVD of medium roots, and RLD of total roots contribute more to the solute transport parameters (Figure 4). Meanwhile, root morphological traits had better explanation on V and D compared with β and ω (Figures 4 and 5). This finding was in agreement with the results obtained by Luo et al. (2019), who found that fine roots would promote the infiltration and rapid solute transport by enhancing preferential flow. This was because, on the one hand, the root volume of fine roots would shrink when the soil was dry, thereby forming root channels at the root-soil interface and promoting preferential flow (Figure 7) (Ghestem et al. 2011; Benard et al. 2021). On the other hand, fine roots often have well-developed shallow

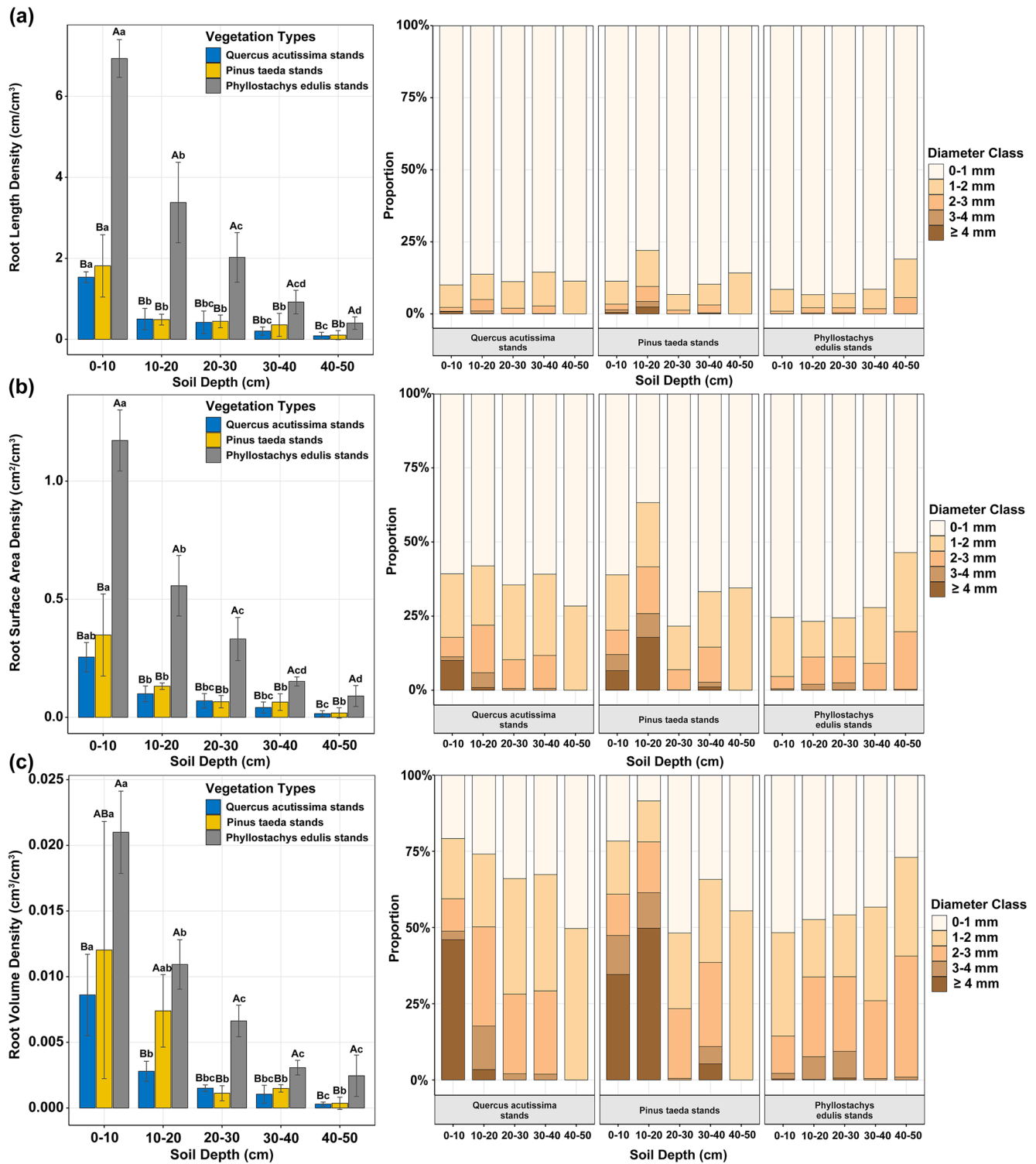


FIGURE 3 | Root morphological traits of the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands). Left panel: The significant differences are the results of the variations in root length density (a), root surface area density (b) and root volume density (c) under different conditions. The different capital letters represent the significant differences in root length density, root surface area density and root volume density of the three forest stands at the same soil depth. The different lowercase letters represent the significant differences in root length density, root surface area density, and root volume density at different soil depths at the same forest stand ($p < 0.05$). Right panel: The proportion of different diameter classes of root systems in all root length density (a), root surface area density (b) and root volume density (c). In this study, roots are classified into four categories, that is, fine roots ($0 < d \leq 1$ mm), medium roots ($1 < d \leq 3$ mm), coarse roots ($3 < d \leq 4$ mm) and extremely coarse roots ($d > 4$ mm).

root systems and high lateral sprouting capacity, which could connect the existing soil pores (Schwärzel et al. 2012; Hoang et al. 2016; Tang et al. 2024b). However, the opposite result

obtained by Cai et al. (2024) was that coarse roots would facilitate preferential flow. They concluded that coarse roots would move the soil particles around the root channel under

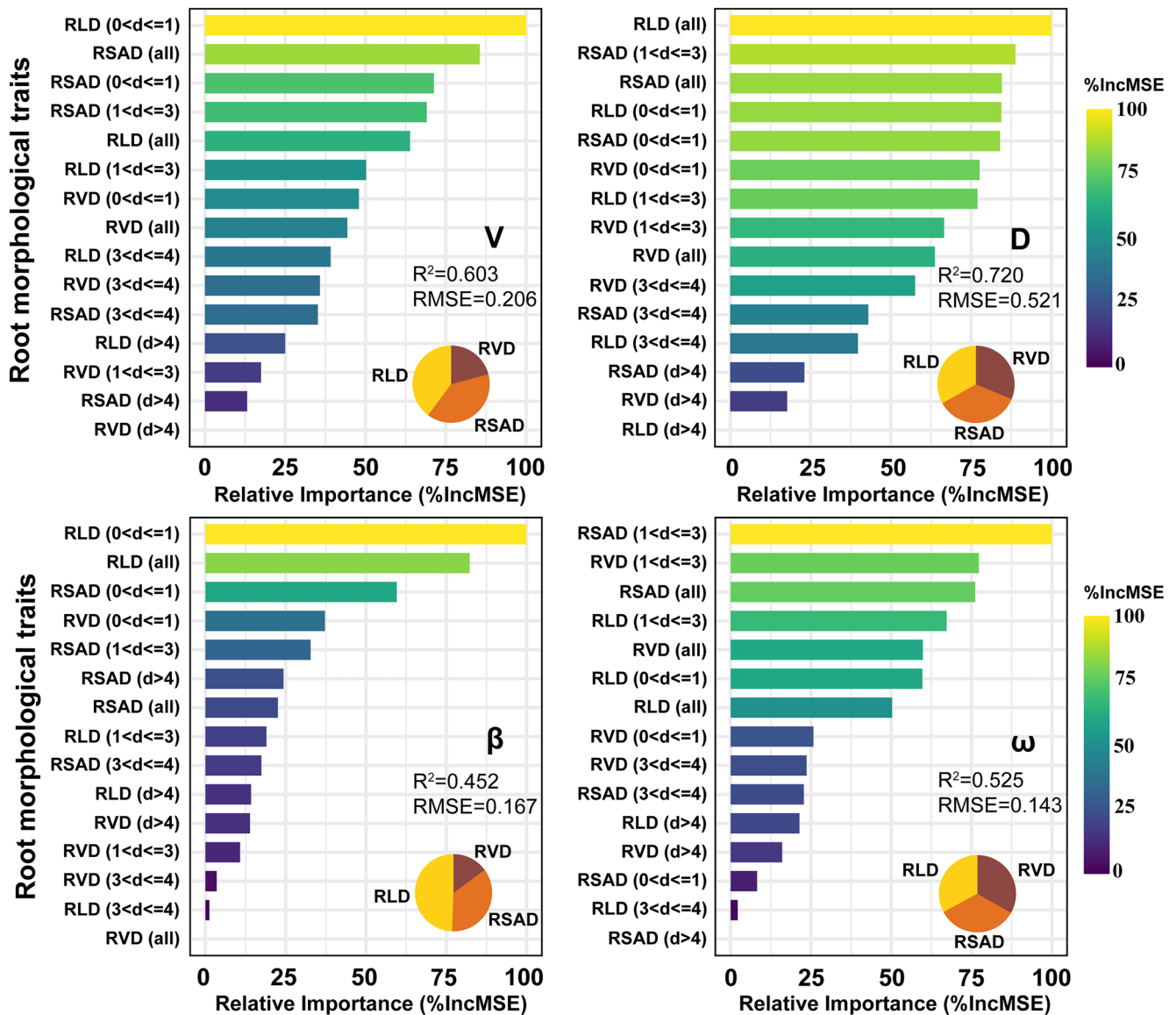


FIGURE 4 | Random forest model (RFM) predicts the importance of root morphological traits (all and for different diameter classes) as drivers for solute transport parameters. Percentage increases in the mean squared error (MSE) of factors were used to estimate the importance of these predictors. The percentage of the predictor with the highest %IncMSE was set at 100%, and the remaining predictors are set as the relative percentages of the predictor with the highest %IncMSE. RLD: Root length density, RSAD: Root surface area density, RVD: Root volume density.

prevailing wind conditions, and thereby, water flows fast along the channel (Guo et al. 2020). A possible reason was that the improving effect of coarse roots on soil infiltration mainly occurs under prevailing wind conditions like the Loess Plateau, while in urban forests, the coarse roots increase the squeezing effect in soil, thus inhibiting the occurrence of preferential flow (Cai et al. 2024). Another possible explanation was that, on the one hand, the Loess Plateau was characterised by high clay content, which is more likely to crack due to the radial pressure exerted by the growth of root systems compared with the silty loam in this study area (Ghestem et al. 2011). Thus, the root-soil interactions were more significant with increasing root diameter. On the other hand, the fine root systems in our study area showed larger root length and stronger branching density compared with those in the Loess Plateau, which further promotes the development of root channels and the connected soil pores (Figure 7) (Hoang et al. 2016). In our

study, we further found that the RLD of the fine roots had the greatest positive contribution on the V, which was demonstrated in the three forest stands (Figures 5 and 6). The RLD could affect the soil water infiltration by influencing the extent of interpenetration and interweaving of the root channel (Liu et al. 2020). In addition, the positive influences of the total RSAD on V should not be ignored (Figures 5 and 6), which was inconsistent with Liu et al. (2020). Biological macropores were formed between the root surface and the surrounding soil matrix, which could act as preferential flow paths and thus have an important role in solute transport (Luo et al. 2019; Liu et al. 2020; Wang et al. 2023). Besides, the root exudates effused from the root surface could improve soil aggregation structure and thus enhance active soil water flow (Carminati et al. 2017; Tang et al. 2024b). Similarly, the RSAD of medium roots and the RLD of the total roots had positive effects on the D (Figures 5 and 6). Theoretically, forest soils

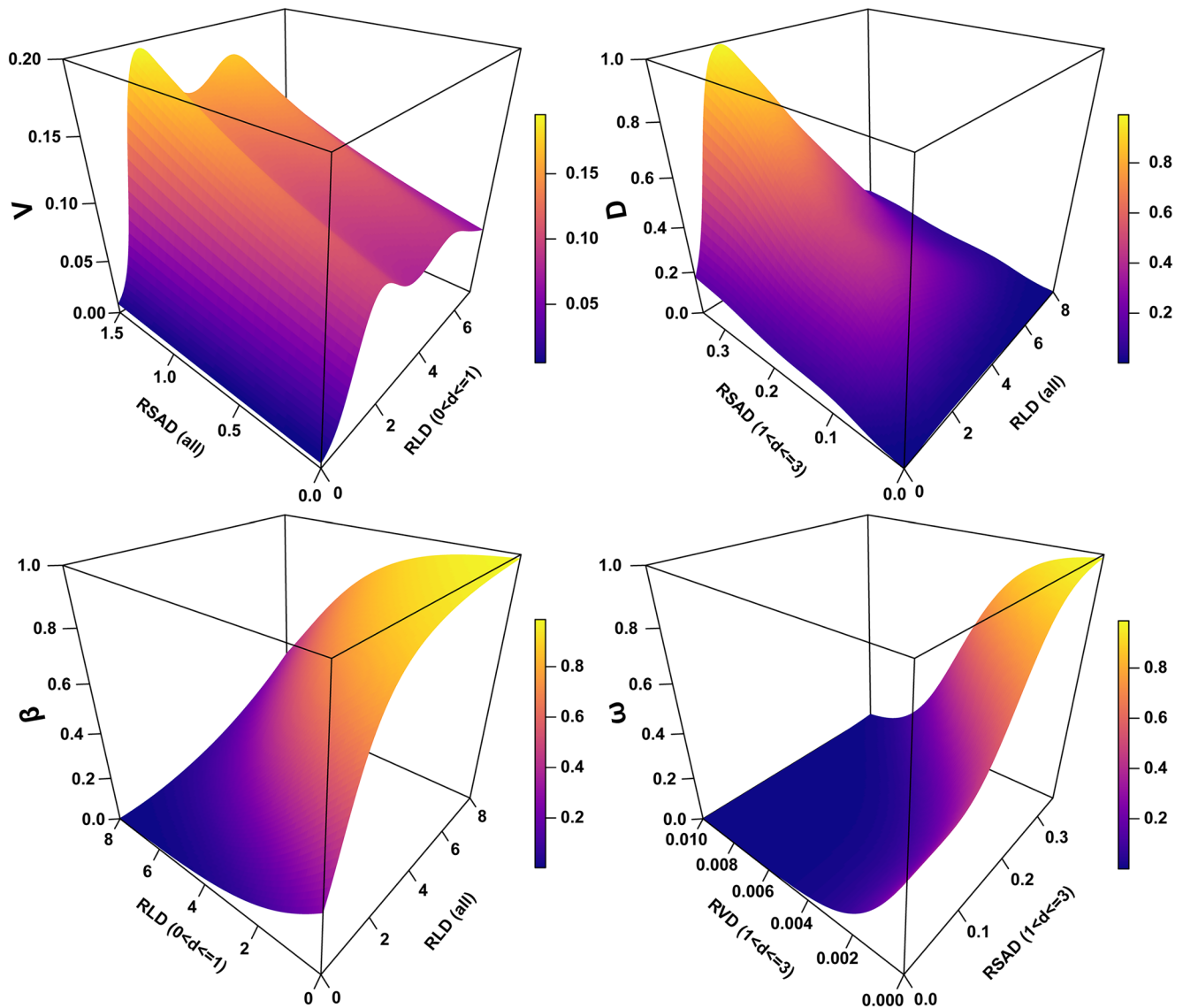


FIGURE 5 | The 3D visualisations of the interactions between solute transport parameters (V , D , β and ω) and the selected root morphological traits, where the colour gradients show low (purple) to high (yellow) values of solute transport parameters. Trends are based on Generalised additive mixed models (GAMMs). RLD: Root length density, RSAD: Root surface area density, RVD: Root volume density.

with well-connected macropores could promote faster and more dispersed solute transport (Ju et al. 2025). Therefore, the D could be enhanced due to the active effects of RLD and RSAD on the connection of the soil pores and the formation of the biological macropores (Figure 7) (Hoang et al. 2016; Wang et al. 2023; Tang et al. 2024b). Although there were no significant relationships between RLD of fine and total roots and β ($p > 0.05$), the positive effects of the RLD of fine and total roots on β should not be neglected (Figures 5 and 6). The penetration of roots into soils could reduce soil compaction and create soil macropores (Sander et al. 2008; Galdos et al. 2020; Zhang et al. 2021b). In addition, the RLD could establish connectivity between existing soil pores and make them more hydrologically active (Figure 7) (Benard et al. 2021). Therefore, the mobile regions would be enhanced. In agreement with Galdos et al. (2020), our results supported that the RVD and RSAD of medium roots had negative effects on the ω . One possible mechanism was that the medium roots could play a similar role as fine roots, like promoting preferential flow to some

extent (Zhang et al. 2021a, 2021b; Chen et al. 2025), which would inhibit the solute diffusion from the mobile to the immobile region (Berkowitz and Scher 2010; Ju et al. 2025). The RVD can promote preferential flow by applying stress to the surrounding soil to create cracks around the roots (Figure 8) (Ghestem et al. 2011; Tang et al. 2024b). Another explanation for this observation was that root exudates secreted from the root surface might create additional capillary forces that reduce the diffusion effects (Carminati et al. 2017).

4.2 | Implication of Forest Management on Soil and Water Conservation

Human activities are relatively frequent around urban forest stands (Xie et al. 2020). In this study area, the rainy season comes from April to June every year, which is also the period for farmers to conduct agricultural production (Tang et al. 2024a, 2024b). Farmers would spray pesticides to control

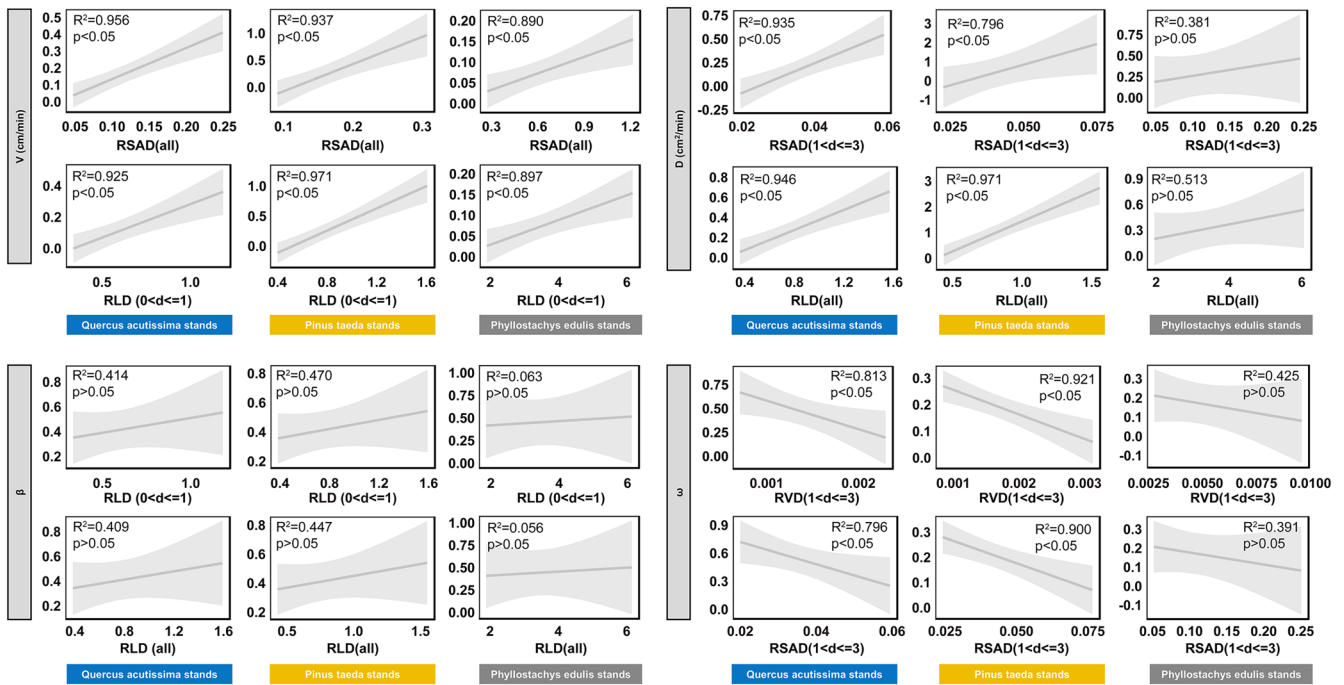


FIGURE 6 | The linear fit between solute transport parameters (V , D , β and ω) and the selected root morphological traits in the three forest stands (*Quercus acutissima* stands, *Pinus taeda* stands and *Phyllostachys edulis* stands). RLD: Root length density, RSAD: Root surface area density, RVD: Root volume density.

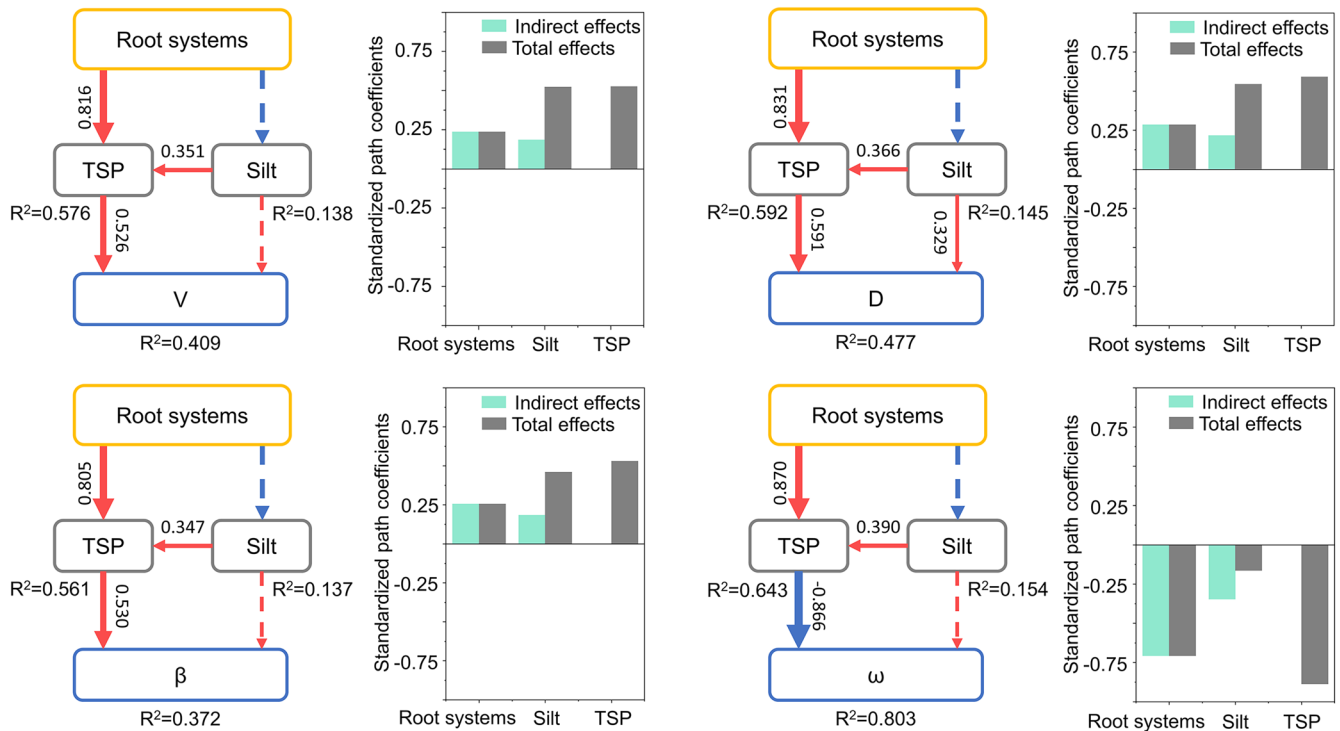


FIGURE 7 | Partial least squares path model (PLS-PM) shows the mechanisms of the root systems, total soil porosity (TSP), and silt content driving the changes in solute transport parameters (V , D , β and ω). The root systems component was the selected root morphological traits obtained from the random forest model.

pests in plantation stands. Therefore, the rainy season is the peak period for the loss of pesticides. The rain would erode the soil, and the loss of pesticides would threaten adjacent stands (Luo et al. 2019; Zhang et al. 2022a). Forests with well-developed soil structures may promote the downward

migration of pollutants and increase the risk of groundwater pollution (Bachmair et al. 2009; Zhang et al. 2015; Luo et al. 2019). In the future, climate change may become more frequent, accompanied by an increase in the frequency of extreme rainfall events (Tang et al. 2024b). Therefore, the

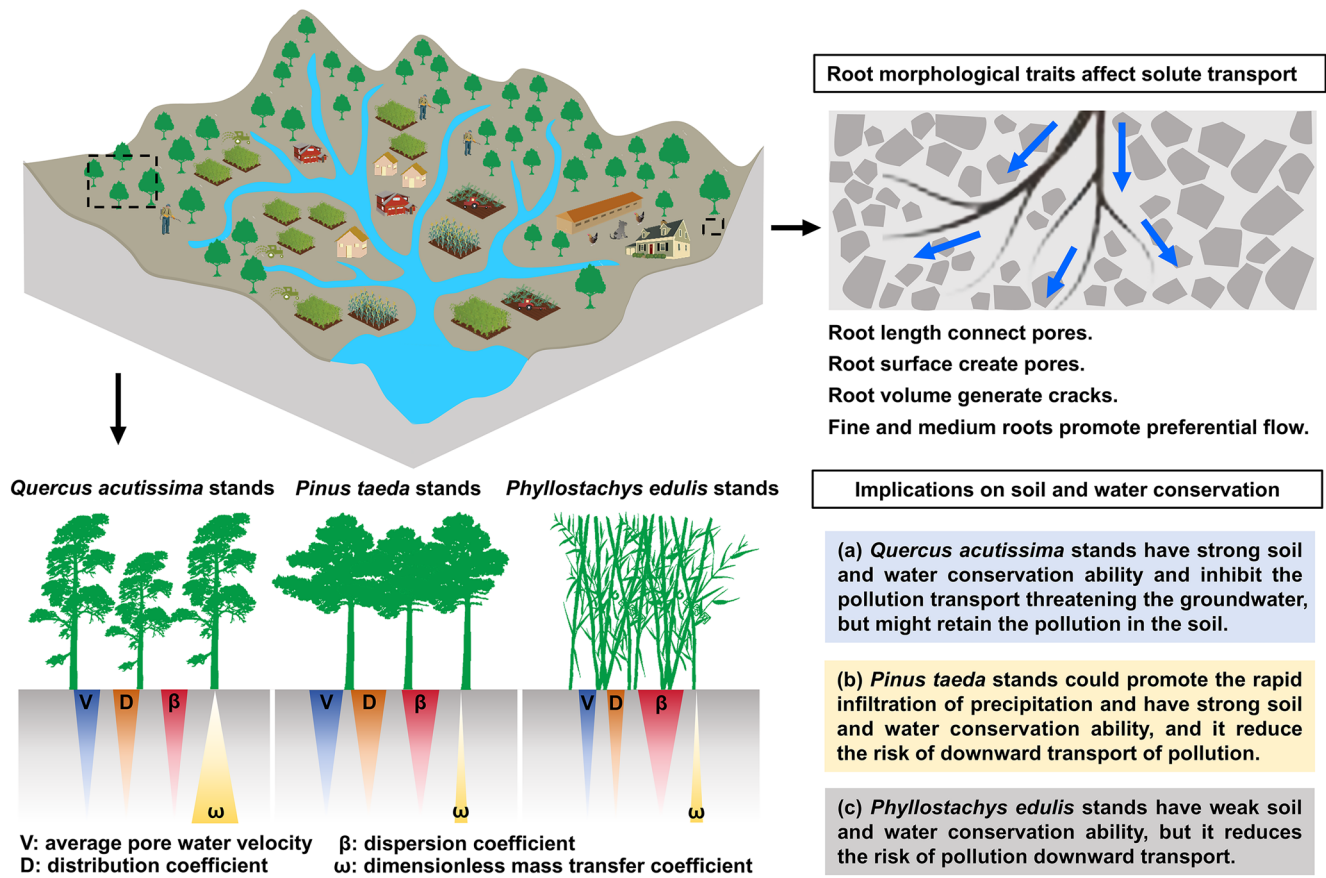


FIGURE 8 | Schematic diagram illustrated that root morphological traits could affect solute transport, and different forest stands could improve watershed water resources management.

management of forest stands is becoming increasingly important. Based on the results of this study, the following suggestions can be provided for forest management:

- I. *Q. acutissima* stands had a larger average pore water velocity and mobile regions in the topsoil (0–20 cm) compared to the other stands, indicating good soil infiltration ability (Table 3). On the contrary, the initial and entire transport time of the subsoil (30–50 cm) is the longest, and the value of average pore flow velocity and the mobile regions are relatively low, indicating that the solute in the subsoil is not easy to infiltrate (Figure 2). Meanwhile, it has the highest mass exchange coefficient ($\omega = 0.44 \pm 0.20$), indicating strong solute exchange ability between the mobile and immobile regions (Table 3). Therefore, *Q. acutissima* stands have relatively strong soil and water conservation and inhibit the solute transport threatening the groundwater. However, the pollution might be retained in the soil due to the strong solute exchange ability between the mobile and immobile regions. According to the actual situation in this area, it can be arranged far away from farmland and plantation stands (Figure 8a).
- II. *Pinus taeda* stands had the highest average pore water velocity, solute dispersion coefficient, and mobile regions in the topsoil (0–20 cm) among the three stands. What's more, its mass exchange coefficient is relatively low (Table 3). Therefore, it is beneficial for the rapid

infiltration of precipitation and has the strongest soil and water conservation compared with the other stands. Meanwhile, it reduces the risk of migration and retention of pollution (Figure 8b).

- III. *Phyllostachys edulis* stands exhibit the lowest average pore velocity and greatest mobile regions across the whole soil profile compared with *Q. acutissima* and *P. taeda* stands, indicating their high connectivity of soil pores but the low vertical solute transport rate (Table 3). At the same time, we found that the litter in the *Phyllostachys edulis* stands after the rainstorm was scoured by runoff, indicating that the soils in *P. edulis* stands were more likely to suffer from raindrop splash erosion (Shinohara et al. 2019). These results demonstrated that *P. edulis* stands have weak soil and water conservation capacity, while they can effectively inhibit the risk of pollution downward transport. Thus, it can be arranged around farmland to act as a barrier against the threat from pesticides (Figure 8c). However, litter erosion might compromise the barrier function of *P. edulis* stands. This is because litter erosion would promote the surface runoff, which in turn increases the pesticide migration (Rajão et al. 2023). Therefore, litter cover management in *P. edulis* stands requires further attention, like introducing more shrubs and herbaceous plants into these stands to improve the litter quality.

4.3 | Limitations and Future Consideration

Previous research showed that roots were the main factor influencing the solute transport (Zhou et al. 2023). This study conducted investigations focusing on different diameter classes of root systems and their morphological traits to the solute transport parameters, which received less attention from current research. However, there are some limitations that need to be considered and can provide guidance for future research. The results demonstrated that the root morphological traits had significant positive effects on solute transport parameters like pore water velocity (V), dispersion coefficient (D) and distribution coefficient (β). This finding was consistent within individual forest stands. Additionally, in comparison with *Q. acutissima* stands, *P. taeda* stands exhibited significantly greater root morphology traits, which were accompanied by larger solute transport parameters (V and D). Nevertheless, although *P. edulis* stands showed the highest root morphology characteristics, they exhibited the highest initial and entire transport time in the topsoil and the smallest solute transport parameters (V and D) in the three forest stands. A possible explanation was that the solute transport process was controlled by root morphological traits and soil properties at the same time (Zhang et al. 2015; Luo et al. 2019; Ju et al. 2025). Therefore, it is required to clarify the interrelationships among the root-soil synthesis and their relative contribution to solute transport (Tang et al. 2024b). What's more, a recent study has found that there was a threshold between root morphological traits and soil water flow behaviour (Qin et al. 2022). Above the threshold of the specific root morphological traits, the soil water behaviour would transfer from soil matrix flow dominated to preferential flow dominated, and the solute transport would be strengthened (Qin et al. 2022). However, the over-dense roots would inhibit fast soil water flow due to the occupation of soil pores (Hao et al. 2021). Thus, whether there is a threshold to facilitate the transfer of preferential flow to soil matrix flow needs to be further paid attention to. In future research, we could set up control experiments to separate root systems and soil effects. Solute transport experiments with different root densities and soil textures could be conducted. In fact, there may be different thresholds of the root morphological traits in soils with various textures. This is because root systems could shape various soil structures and soil pore densities based on the soil textures (Shi et al. 2021). In addition, to better capture the impact of root-soil interactions on solute transport, X-ray computed tomography (CT) can be used to record three-dimensional images and identify the spatial distribution of soil water in macropores (Chaif et al. 2025).

5 | Conclusion

In this study, the authors tried to answer the hypothesis that root morphological traits in different diameter classes have various contributions on the solute transport. Several key findings were yielded as follows: (1) Root morphological traits and solute transport patterns vary in different forest stands. *Phyllostachys edulis* stands showed significantly greater RLD and RSAD than those in both *Q. acutissima* and *P. taeda* stands. *Phyllostachys edulis* stands showed the slowest initial transport in topsoil, whereas *Q. acutissima* stands demonstrated the slowest initial and entire transport in subsoil. (2) Different root morphological traits exert specific contributions to solute transport in the

soil. Fine root RLD and total RSAD promoted solute velocity, medium root RSAD and total root RLD enhanced the dispersion coefficient, and both medium root RLD and total root RLD facilitated the mass transfer coefficient. Conversely, medium root RSAD and RVD negatively impacted the mobile water fraction. (3) Different forest stands showed various significance in practice based on the solute transport processes. Forest managers could increase the proportion of *P. taeda* stands to enhance soil and water conservation. For the simple purpose of reducing the risk of pollutant downward transport, establishing *P. edulis* stands around farmland might be a viable practice.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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